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Kelas : TI-24-KA

Matkul : Lab. Pemrograman Web

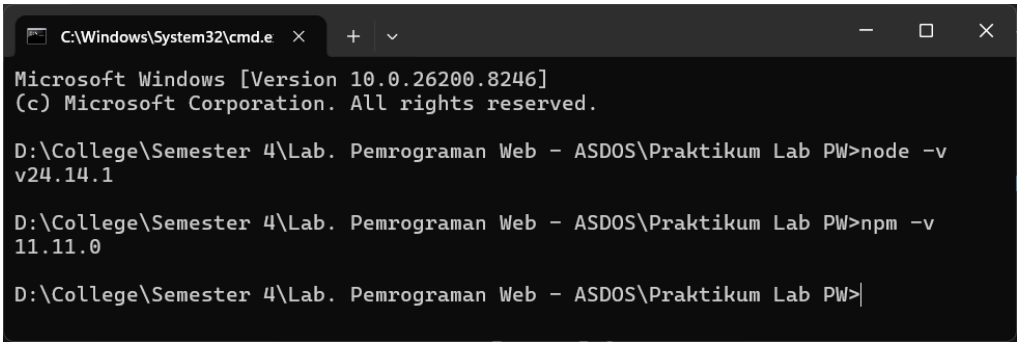
Pertemuan : 4

Tugas dan Latihan

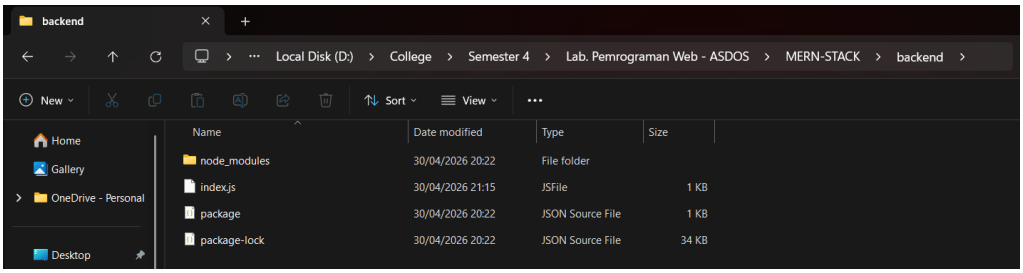
Praktikum ini bertujuan memahami konsep full-stack development menggunakan Node.js, Express, React, dan MySQL serta implementasi REST API sederhana.

Langkah Pengerjaan

1. Setup Backend
   A. Screenshot install Node.js



   B. Screenshot folder backend



C. Screenshot install

```
C:\Windows\System32\cmd.e

D:\College\Semester 4\Lab. Pemrograman Web - ASDOS\Praktikum Lab PW>npm init -y
Wrote to D:\College\Semester 4\Lab. Pemrograman Web - ASDOS\Praktikum Lab PW\package.json:

{
  "name": "praktikum-lab-pw",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "repository": {
    "type": "git",
    "url": "git+https://github.com/raflymy47/PW-Praktikum-242310014.git"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "bugs": {
    "url": "https://github.com/raflymy47/PW-Praktikum-242310014/issues"
  },
  "homepage": "https://github.com/raflymy47/PW-Praktikum-242310014#readme"
}


D:\College\Semester 4\Lab. Pemrograman Web - ASDOS\Praktikum Lab PW>
```

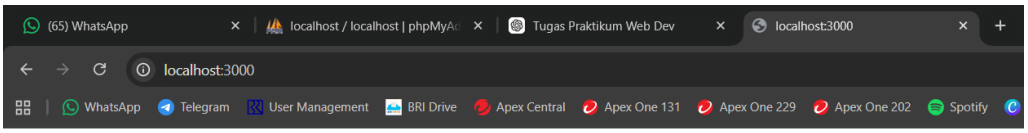
```
C:\Windows\System32\cmd.e

D:\College\Semester 4\Lab. Pemrograman Web - ASDOS\Praktikum Lab PW>npm install express mysql2 cors

added 78 packages, and audited 79 packages in 6s

25 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

D:\College\Semester 4\Lab. Pemrograman Web - ASDOS\Praktikum Lab PW>
```

2. Membuat Server Express

A. Screenshot file index.js

```
index.js
File   Edit   View

const express = require('express');
const mysql2 = require('mysql2');
const cors = require('cors');
// const { version } = require('react');

const app = express ();

app.use(express.json());

app.use(cors({
    origin: 'http://localhost:3000',
    credentials : true
}));

const db = mysql2.createConnection({
    host: 'localhost',
    user: 'root',
    password: '',
    database: 'pw_praktikum_db'
});

db.connect((err) => {
    if (err) {
        console.error('Error connecting to database:', err);
        return;
    }
    console.log('Connected to MySQL database');
});

app.get("/", (req, res) => {
    res.send(" Server bejalan dengan baik");
});

app.listen(3001, () => {
    console.log("Server running on port 3001");
});

app.get("/api/info", (req, res) => {
    res.json({
        message: "API MERN Stack Build by Express JS",
        version: "v1.0.0",
        status: "active",
        timestamp: new Date()
    });
});
```

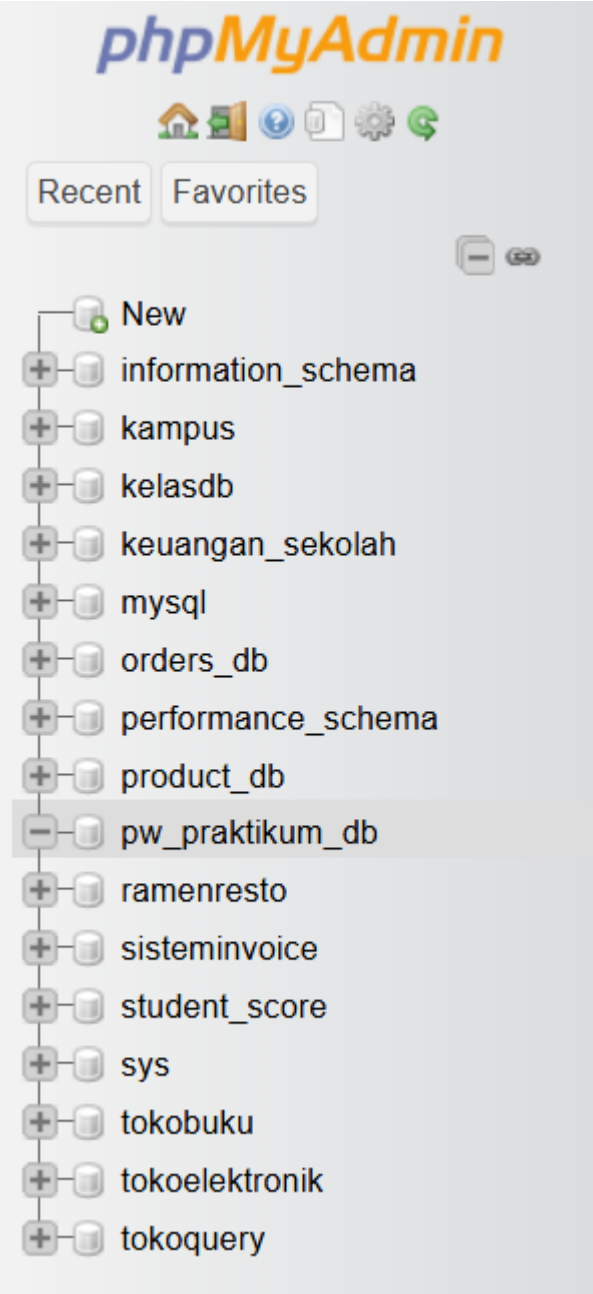
B. Screenshot hasil di browser: localhost:3000



Server bejalan dengan baik

3. Koneksi Database MySQL

A. Screenshot database pw_praktikum_db



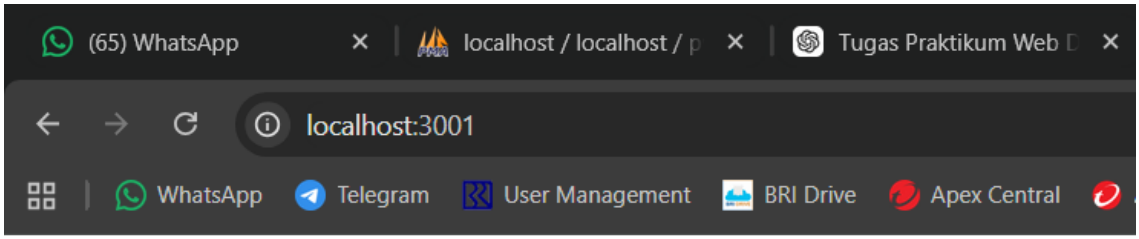
B. Screenshot kode koneksi MySQL

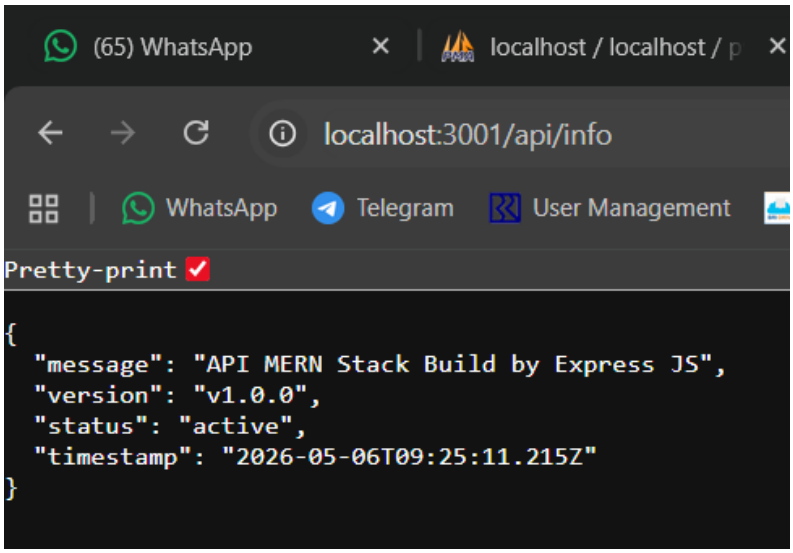
```
const db = mysql2.createConnection({
    host: 'localhost',
    user: 'root',
    password: '',
    database: 'pw_praktikum_db'
});

db.connect((err) => {
    if (err) {
        console.error('Error connecting to database:', err);
        return;
    }
    console.log('Connected to MySQL database');
});

```

C. Screenshot hasil koneksi berhasil

```
C:\Windows\System32\cmd.e

D:\College\Semester 4\Lab. Pemrograman Web - ASDOS\MERN-STACK\backend>node index
Server running on port 3000
Connected to MySQL database
```

4. Setup Frontend React



Server bejalan dengan baik

5. Integrasi Backend & Frontend